ISyE 6339 Physical Internet Engineering

Caseworks 2.1 and 2.2

Designing Hyperconnected Multi-Tier Megaregional Freight Systems

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Due time: 2026/04/30

Network-centric CW 2.1: 40 points
Hub-centric CW 2.2: 20 points

Teams must select between performing only CW 2.1 or performing both caseworks 2.1 & 2.2

Team size: 2-4 students
Casework grading does not account for team size

Teams must provide their generated code and their computational outcome files

This casework set focuses on *designing a scalable hyperconnected multi-tier megaregional freight system* for one of the following megaregions:

- **Northeast USA:** *West Virginia, Virginia, Maryland, Delaware, Washington D.C., Pennsylvania, New York, New Jersey, Connecticut, Massachusetts, New Hampshire, Vermont, Maine, Rhode Island*
- **Southwest USA:** *Texas, Oklahoma, Arizona, New Mexico*
- **Midwest USA:** *Kansas, Missouri, Illinois, Ohio, Indiana, Michigan, Kentucky, Iowa, North Dakota, South Dakota, Nebraska, Minnesota, Wisconsin*
- **West USA:** *Nevada, Utah, Colorado, Wyoming, Montana, Idaho, California, Oregon, Washington*

The emphasis here is on building a realistic and scalable baseline design using the same pragmatic strategies illustrated in the tutorial in-class presentation: demand-informed spatial structuring, realistic hub candidate screening, set-cover based hub filtering, and network construction across multiple tiers.

Each team will be assigned a specific megaregion other than the Southeast USA used in the tutorial. The mechanism used for selection is as follows: first come first choice, with a second team in a megaregion allowed only once all have been assigned to a team. Praveen is to be contacted on Teams to request a megaregion.

Students must interpret the geography, freight structure and logistics infrastructure of that megaregion, then propose a scalable design for regional hubs and gateway hubs, with clear reasoning on how the megaregion interfaces with the rest of the country via global hubs and other megaregions via interregional hubs.

The objective is to produce a defensible and reproducible first-pass multi-tier design that could plausibly guide future discussions with regional stakeholders, infrastructure owners, and logistics operators. A pragmatic, scalable, and well-justified baseline is valued more than a highly complex model with weak interpretability.

Data Sources:

The tutorial emphasizes practical design using publicly available freight data and realistic logistics asset data. The casework is expected to rely primarily on the following sources:

1. **Freight Data: Freight Analysis Framework (FAF)**
   The casework relies on statistics and forecasts compiled by the Bureau of Transportation Statistics (BTS) within its Freight Analysis Framework (FAF). We suggest students to check out FAF website FAF and read carefully the FAF User Guide.

a. **Freight Regional Database**
   FAF_Regional_Database provides origin-destination freight transportation demand between 132 FAF zones, aggregated by 3 trade types, 8 transportation modes, 42 commodity types and multiple distance bands. The lists of each category can be found at the FAF User Guide and FAF5_metadata.xlsx in the zip file.

b. **Experimental County-to-County Estimates**

   The Bureau of Transportation Statistics (BTS) developed an experimental county-to-county commodity flow product to provide the user community with more geographically granular commodity flow data to support planning, policymaking, and operational decisions at the state and local levels. We suggest students to study the user guide for FAF County Level Estimates at http://www.bts.gov/faf/county/documentation

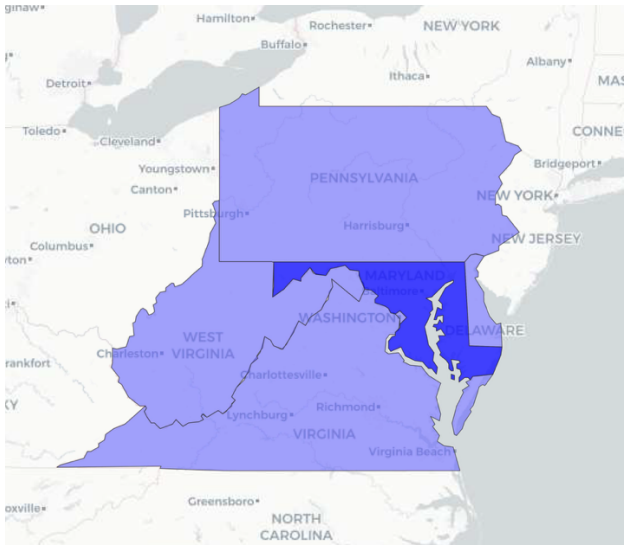
   One can download two kinds of files: state-specific files and disaggregation factors for all counties in the U.S. In the FAF Regional Database, demand was aggregated to 42 commodity groups and 132 FAF Zones. Now, these commodity types are clustered into five commodity groups as shown in the table below:

sctgG5 Code	Definition	FAF SCTG Code
sctg0109	Agricultural products	1-9
sctg1014	Gravel and mining products	10-14
sctg1519	Coal and other energy products	15-19
sctg2033	Chemical, wood and metals	20-33
sctg3499	Manufactured goods, mixed freight, waste and unknown	34-99

The following table lists the mode groups and their codes in the new database

New Mode Code	New Mode Definition	FAF Mode Code
11	Truck and Air	1 – Truck 4 – Air
2	Rail	2 – Rail
3	Water	3 – Water
5	Multiple modes and mail	5 – Multiple modes and mail
6	Pipeline	6 – Pipeline

The release included 51 state-specific zip files. Each file represents flows using county-level geography for the main state and states adjacent to the main state, together termed as "Multi-State". For example, Figure below shows the multi-state area of Maryland and its immediate area, which includes four adjacent states (Pennyslvania, Delaware, Virginia, and West Virginia) and the District of Columbia.



Source : BTS and OpenStreetMap 2024.

The full set of county-level factors (origin and destination factors) are contained in one zip file All_Experimental_Disaggregation_Factors.zip. One can merge these factors with FAF regional database to create a county-level database for a customized geographic area or for the entire U.S. The factors are available for four modes and five commodity groups:

- Rail (rail_origin_factors.csv and rail_destination_factors.csv)
- Water (water_origin_factors.csv and water_destination_factors.csv)
- Truck (truck_origin_factors.csv and truck_destination_factors.csv)
- Pipeline (pipeline_origin_factors.csv and pipeline_destination_factors.csv)

One can apply the truck factors to air and multiple modes and mail modes to generate county-to-county flows for these modes. This experimental product does not include methods to disaggregate flows by other or unknown mode or flows that use no domestic mode.

Applying Disaggregation Factors:

1. Select FAF trips for the area of interest. For example, to generate county-to-county tonnage estimates only for Iowa to Idaho, the user can select flows with *dms_orig* = 190 and *dms_dest* = 160, filtering out all other flows
2. Select FAF trips for the mode of interest. For example, to disaggregate rail flows, select flows with *dms_mode* = 2 (rail) from the previous step, filtering out all other flows
3. Use the commodity correspondence from Table to summarize flows using the five commodity groups. This will reduce the 42 commodity categories to five.
4. Join the resulting table from the previous step to the origin factor table based on the FAF *origin_zone (dms_orig)* and commodity group (*sctgG5*) columns. The resulting file now contains the factor for the origin county and commodity group *(f_orig)*.
5. Join the resulting table from the previous step to the destination factor table based on the FAF destination zone (*dms_dest*) and commodity group *(sctgG5)* columns. The resulting file now contains the factor for the destination county and commodity group *(f_dest)*.
6. The resulting table contains all possible combinations of county pairs and commodity groups that are present in the original FAF estimates for the selected area and mode. To calculate the final county-to-county tonnage by commodity group, multiply the total FAF zone-level tons by the origin factor and the destination factor.

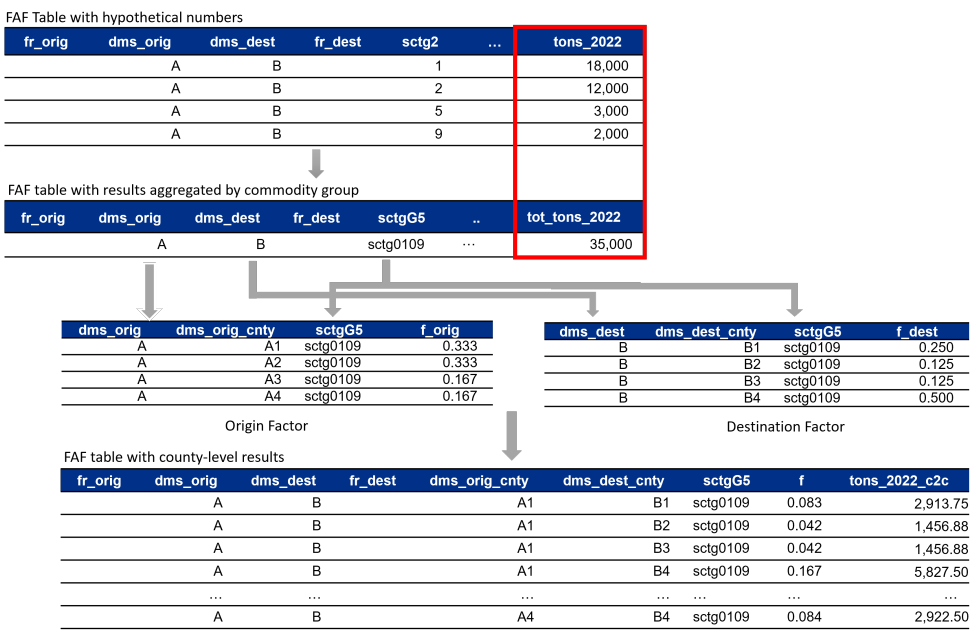
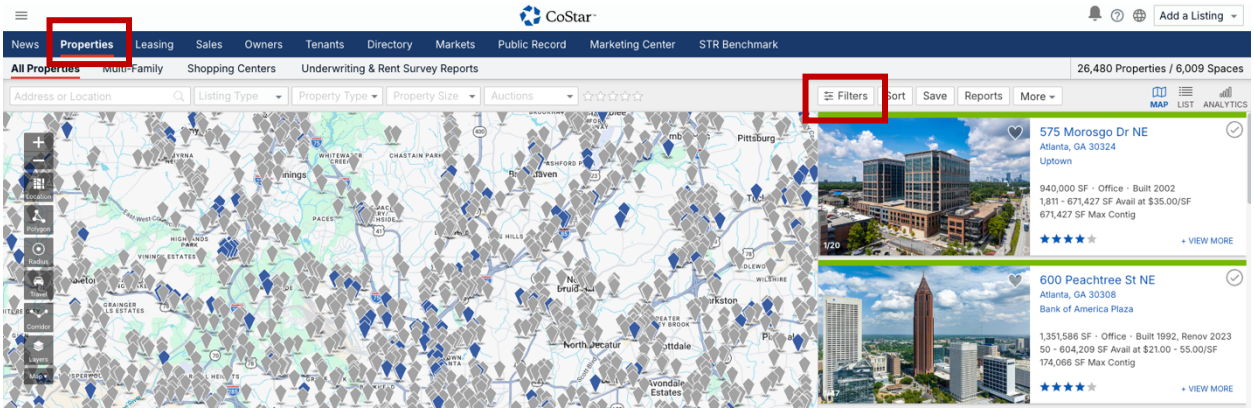


Figure 1: Procedure to obtain county level flows from FAF regional flows

2. Commercial real estate / logistics facility data: CoStar

CoStar https://www.costar.com/) is a commercial real estate intelligence platform that provides detailed, property-level data on building assets across the globe. It includes information such as property location, space use, available size, land area and tenant mix. In this casework, CoStar can be used to identify proxies for hub candidates by filtering Properties based on criteria such as space use (e.g. industrial, retail etc.), secondary type (e.g. distribution center, warehouses, truck terminals etc.) and available space (e.g. 200,000 – 500,000 sq ft.)



3. Major Infrastructure Layers:

Interstates, major U.S. and state routes (NorthAmericanRoads) where relevant, railroads, ports, cargo airports, border crossings, and intermodal terminals / railyards should be used to interpret freight structure. Some of these are given as part of the working folder while others can be obtained via Data Catalog.

Tasks for Network-Centric Casework 2.1:

1. Construct the county-to-county freight flow matrix for the megaregion based on the given procedure. Identify major inbound, outbound, and internal flows, and present pareto-style summaries across counties and county groupings by mode, trade type and commodity groups, for 2025 and 2030. Analyze your results, aiming to provide key insights.

2. Identify the major global, continental and national interface nodes relevant to the megaregion, such as seaports, cargo airports, border crossings or external demand-dense hubs. For clarity and emphasis, you are encouraged to produce distinct maps showing the expected volume flowing through these nodes.

3. Cluster the counties of the megaregion into a set of regions:
    a. Develop a clustering approach that aims to balance freight demand while preserving geographic contiguity and compactness, and that reflects the influence of major transportation infrastructure such as interstate corridors and railroads.
    b. Clearly explain and justify your clustering logic, parameters, and assumptions.
    c. Present the resulting regions and assess how well they perform with respect to freight balance, compactness, contiguity, and infrastructure alignment. Identify any regions that appear problematic or difficult to justify and discuss whether further refinement would be warranted.

4. Using the provided CoStar data and any other relevant logistics asset information:
    a. Identify the set of feasible candidate locations for regional hubs within the megaregion. Explain the filters, thresholds, and practical assumptions used to screen the property-level dataset.
    b. Distinguish between existing facilities that appear directly usable and locations that should be interpreted only as proxies for feasible hub deployment.

5. Develop and apply a methodology to filter regional hub candidates and define the regional hub network for the megaregion.
    a. Formulate and implement a justified filtering approach, such as a set-cover-based optimization model, heuristic, or other scalable method, to reduce the initial set of feasible regional hub candidates. Ensure that the proposed methodology accounts for regional coverage requirements, accessibility to major transportation infrastructure, realistic facility availability, and proximity to the freight concentration of the region.
    b. Based on the filtered candidate set, identify the regional hubs to be retained in the proposed network and, for each selected hub, specify its location, the region or regions it primarily serves, and the rationale for its inclusion.

c. Define the principal regional hub network links between the selected hubs, using travel time, infrastructure continuity, and expected freight interaction as key considerations.
d. Present maps of the resulting regional hub network and analyze its main structural characteristics and implications.

6. Develop the lower-tier territorial structure and gateway hub design for the megaregion.
    a. Within each generated region, group counties into freight areas (urban or subregional) while respecting the established regional boundaries and reflecting freight intensity, metropolitan concentration, and accessibility through major highways and logistics infrastructure.
    b. Using the provided facility dataset, identify and screen candidate gateway hub locations within these urban areas based on realistic logistics land use, approximate facility size, accessibility, and expected functional role.
    c. Develop and apply a suitable methodology to select the gateway hubs so that the areas are meaningfully connected to the regional tier while avoiding excessive redundancy or unrealistic deployment.
    d. Present the selected gateway hubs on appropriate maps and analyze how they are distributed relative to metropolitan concentrations, regional hubs, and major infrastructure.
    e. Explain and justify the overall lower-tier design methodology and its main outcomes.

7. Integrate the selected regional hubs, gateway hubs, regions, and urban areas into a single multi-tier hyperconnected network design for the megaregion.
    a. Depict the resulting hierarchy through maps and diagrams.
    b. Explain how the proposed structure supports freight movement from global and national interfaces into the megaregion, across the megaregion, and toward major urban concentrations.
    c. Highlight the principal structural characteristics of the proposed design.

8. Using reasonable simplifying assumptions, assign freight flows to the proposed multi-tier network.
    a. Estimate the major flows between interface nodes, regional hubs, gateway hubs, and their served territories for 2025 and 2030.
    b. Highlight the most critical hubs, the highest-throughput corridors, and the strongest freight concentration patterns emerging from your design.

    Analyze your results, aiming to provide key insights.

9. Synthesize the principal challenges, insights, and lessons learned from designing a scalable hyperconnected multi-tier regional freight network for the assigned megaregion. Notably:
    a. Reflect on the limitations of the available data.
    b. Discuss the strengths and weaknesses of the adopted methodology.
    c. Identify the most important next steps for refining the design in future work.

Tasks for Hub-Centric Casework 2.2:

The tasks of hub-centric casework 2.2 are meant to be realized concurrently while progressing through network-centric casework 2.1.

1. Develop alternate physical design concepts for logistic hubs at the principal tiers of the proposed network and characterize their expected operations.
    a. For each major hub tier considered in your design, at minimum including regional hubs and gateway hubs, develop alternative physical design concepts that could realistically support their expected role within the megaregion. Where relevant, you may also include conceptual designs for major interface hubs connected to ports, airports, rail terminals, or border crossings.
    b. For each alternative design, describe the principal operating functions it is intended to support, such as freight consolidation and deconsolidation, cross-docking, relay and transfer operations, trailer or container staging, temporary buffering, re-energization, vehicle dispatching, and other relevant value-adding or support activities.
    c. Compare the alternative designs in terms of land footprint, facility layout, level of automation, vehicle interface, throughput orientation, storage intensity, scalability, and suitability for different freight profiles or corridor roles.
    d. Provide diagrams, layout sketches, or structured illustrations to support your proposed designs, and discuss which alternative appears most appropriate for the principal hubs in your network.

2. Synthesize the principal challenges, insights, and lessons learned from designing representative logistic hubs for a scalable hyperconnected multi-tier regional freight network for the assigned megaregion. Notably:
    a. Reflect on the limitations of the available data.
    b. Discuss the strengths and weaknesses of your adopted methodology.
    c. Identify the most important next steps for refining the designs in future work.

We hope these caseworks prove to be a challenging yet highly valuable learning experience

We recommend you start early and coordinate your work among team members.